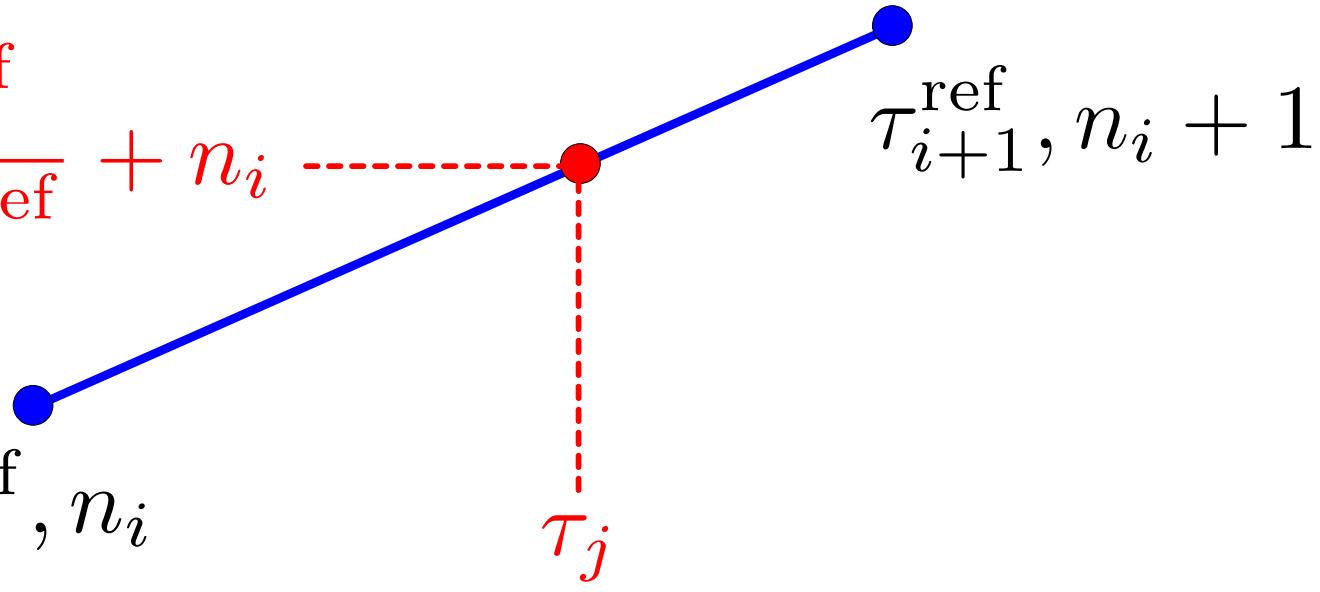$$n'_j = \frac{\tau_j - \tau_i^{\mathrm{ref}}}{\tau_{i+1}^{\mathrm{ref}} - \tau_i^{\mathrm{ref}}} + n_i$$